

Accuracy and Significant Digits

Using a calculator does *not* guarantee accuracy. To obtain the right answer, you need to follow the rules for rounding based on *significant digits*.

Just because the calculator or software will show a result with 10 decimal places does not mean you should report these.

Do not round intermediate calculations. Only round off the final answer. For example, if you round-off *any* intermediate result to two digits, you discard the information so your final answer might be judged incorrect (“round-off error”)

Here are six basic rules. *In statistics, Rule 6 is especially important.*

<i>Rule</i>	<i>Examples</i>	<i>Significant Digits</i>
Rule 1. Non-zero digits (1-9) are always significant.	875 875.4 1.256E+06	Three Four Four (this is 1.256×10^6)
Rule 2. Zeroes <i>between</i> other digits are significant	609 12002	Three Five
Rule 3. Zeroes <i>after</i> other digits but <i>after</i> a decimal point are significant	7.40 0.013000	Three Six
Rule 4. Zeroes at the <i>end</i> of a number are significant only if they are <i>after</i> a decimal point. Otherwise, you can’t be sure (the data may have been rounded). Look at the rest of the numbers in the data column for clues.	5700.0 5700	Five At least two (if rounded to nearest 100) but could be three or four. Check other numbers in the list (if you see a number like 6507 assume four-digit accuracy).
Rule 5. Zeroes <i>before</i> other digits may not be significant (but look at the rest of the data in the data column for clues).	0.058	Two (but if the next number is 0.174 then assume three-digit accuracy).
Rule 6. The number of significant digits is limited to the <i>least</i> significant digits in any numbers in a product or quotient.	(12.4)/(8.4077) 12.37×5.6	Three Two